Reality Abstraction Module (RAM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™
Architecture Family: SIMULOS® — Simulation Governance Architecture
Parent Standard: Reality Representation Standard (RRS)
Operational Layer: Simulation Reality Representation Layer
Category: Governance & Enforcement
Subcategory: Simulation Governance
Type: Parent Standard Module
Governed Space: Reality Abstraction
Version: 1.0
Status: Canonical · Open Module
Effective Date: 16 August 2026
Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS® · INTEGROS® · ADIT® · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™
AI-Readable: Yes
Authority: OOF®
Protection: MIP® — Methodological Intellectual Property
Canonical Language: English (UCL™)

Governed Space

Reality Abstraction

Canonical Definition

Reality Abstraction Module (RAM) defines the governance conditions under which Reality Elements selected as materially relevant to a simulation are simplified, generalized, aggregated, decomposed, parameterized, discretized, approximated, or otherwise transformed into representations suitable for simulation while preserving the characteristics necessary to answer the governed simulation inquiry.

RAM establishes the governance boundary between selected reality and its simulation-level representation, ensuring that reduction of real-world complexity remains explicit, justified, traceable, and bounded by Simulation Purpose & Scope.

Operational Role

RSM establishes:

What parts of reality matter to the simulation?

RAM asks the next question:

At what level of abstraction may those selected parts of reality be represented without removing characteristics material to the simulation inquiry?

The governed progression is:

Reality Selection Basis → Reality Element → Material Characteristics → Abstraction Requirement → Abstraction Decision → Abstraction Limitation → Abstracted Reality Representation → Representation Fidelity

RAM does not determine the required fidelity or resolution of the completed representation.

It governs how real-world complexity may legitimately be reduced before those subsequent representation properties are established.

Module Operational Space

RAM governs: abstraction objectives, abstraction levels, abstraction boundaries, material characteristic preservation, simplification, generalization, aggregation, decomposition, parameterization, discretization, approximation, behavioral abstraction, structural abstraction, environmental abstraction, temporal abstraction, spatial abstraction, interaction abstraction, dependency abstraction, state abstraction, process abstraction, actor abstraction, abstraction assumptions, abstraction justification, abstraction limitations, abstraction uncertainty, abstraction sensitivity, abstraction consistency, abstraction completeness, abstraction traceability, abstraction change, and Abstraction Basis governance.

Module Function

Reality contains more complexity than most simulations can or should reproduce.

A governed simulation therefore requires abstraction.

Abstraction itself is not a defect.

Uncontrolled abstraction is.

RAM exists to prevent: arbitrary simplification, convenience-driven abstraction, abstraction based solely on computational limitations, removal of materially relevant characteristics, excessive aggregation, hidden assumptions, loss of important dependencies, elimination of relevant failure behavior, unrealistic behavioral simplification, abstraction inconsistent with Simulation Purpose, inconsistent abstraction across related simulation components, and representation of an abstracted model as if it preserved characteristics that were intentionally removed.

Reality Abstraction Principle

SIMULOS® establishes:

Reality may be simplified only to the extent that the simplification does not remove characteristics materially necessary to the governed simulation inquiry.

The objective is not:

maximum realism.

The objective is:

sufficient representation of material reality for the intended simulation purpose.

Abstraction Is a Governed Transformation

Reality Abstraction transforms:

Selected Reality

into:

Abstracted Reality Representation

This transformation should remain reconstructable.

For every material abstraction, it should be possible to understand:

What existed in Relevant Reality?

What was selected?

What characteristics were retained?

What characteristics were simplified or removed?

Why was that abstraction considered acceptable?

What limitations does the abstraction create?

Abstraction Object

An Abstraction Object is the Reality Element, relationship, behavior, process, environment, condition, or other selected reality component undergoing abstraction.

An Abstraction Object may include: a physical entity, a human actor, an autonomous actor, a process, an environment, a dependency, an interaction, a failure mechanism, a temporal process, a spatial relationship, or another selected Reality Element.

Material Characteristic

A Material Characteristic is a property of a selected Reality Element whose removal, distortion, aggregation, or simplification could materially affect the simulation inquiry.

Material Characteristics may include: physical properties, behavior, state transitions, response characteristics, dependencies, constraints, timing, geometry, variability, uncertainty, interaction patterns, failure behavior, or environmental sensitivity.

RAM requires these characteristics to be identified before significant abstraction occurs.

Abstraction Requirement

An Abstraction Requirement defines what characteristics must remain represented after abstraction.

It connects:

Reality Selection

to:

Representation Design

and prevents important characteristics from disappearing silently during model simplification.

Abstraction Level

An Abstraction Level describes the degree to which a Reality Element is represented with reduced complexity.

Conceptually, representation may range from:

Detailed Representation

through:

Reduced Representation

Aggregated Representation

Parameterized Representation

to:

Highly Abstract Representation

These are methodological descriptions rather than universal technical classes.

SIMULOS® does not prescribe one abstraction scale for all simulation domains.

Abstraction Must Be Purpose-Relative

There is no universally correct abstraction level.

A representation that is sufficient for one simulation may be inadequate for another.

For example, a pedestrian may be represented as:

a moving point

for one traffic-flow simulation,

but may require:

orientation, reaction time, visibility, movement variability, decision behavior, and interaction characteristics

for an autonomous-vehicle safety simulation.

The correct abstraction depends upon what the simulation must legitimately examine. Simplification

Simplification removes unnecessary complexity while retaining characteristics required by the simulation inquiry.

A simplification should be distinguishable from an omission.

RSM determines whether the Reality Element belongs in the representation.

RAM determines which characteristics of that selected element may be simplified. Generalization

Generalization represents multiple detailed real-world characteristics through a broader representation.

Generalization may be appropriate where detailed differences are not material to the simulation inquiry.

Where differences could materially affect outcomes, generalization requires stronger justification or should not be used. Aggregation

Aggregation combines multiple Reality Elements, states, behaviors, events, or properties into a collective representation.

Aggregation may reduce computational and structural complexity.

However, it may also hide: local variation, minority behavior, extreme conditions, nonlinear interactions, individual failures, dependency structure, or rare but consequential states.

RAM therefore requires material aggregation effects to remain visible. Decomposition

Abstraction does not always mean combining reality.

A Reality Element may need to be decomposed into smaller representational components where its internal structure materially affects simulation behavior.

RAM governs whether such decomposition is necessary to preserve the characteristics relevant to the inquiry. Parameterization

Parameterization represents complex real-world behavior or characteristics through governed parameters.

Parameterization should preserve: parameter meaning, applicable conditions, source basis, assumptions, ranges, and known limitations

where these materially affect representation. Discretization

Continuous real-world phenomena may be represented through discrete: time steps, spatial cells, states, categories, events, or value ranges.

RAM governs the methodological decision to discretize.

The specific granularity required is subsequently governed through Representation Resolution. Approximation

Approximation may be necessary where exact representation is impossible, unnecessary, or computationally disproportionate.

An approximation should remain bounded by: the simulation inquiry, material characteristics, expected behavior, known error, applicable conditions, and downstream interpretation.

Behavioral Abstraction

Real-world actors may exhibit highly complex behavior.

RAM may govern abstraction of: human behavior, autonomous behavior, organizational behavior, agent behavior, biological behavior, or system response.

Behavioral abstraction must not silently eliminate behavior that is material to the simulation inquiry.

Human Abstraction

Humans may be represented as: deterministic actors, probabilistic actors, behavioral classes, population distributions, decision rules, or more complex cognitive representations.

RAM governs whether the selected abstraction is appropriate to the simulation inquiry.

It does not govern human cognition itself.

Where cognition requires independent governance, CLIA® provides the relevant complementary architecture.

Autonomous Actor Abstraction

Autonomous actors may require abstraction of: perception, reasoning, decision behavior, permissions, constraints, coordination, escalation, adaptation, or response.

RAM governs only how these characteristics are represented inside the simulation.

Governance of the autonomous system itself remains within ASGA™ and other applicable architectures.

Structural Abstraction

Complex systems may be structurally simplified.

Structural abstraction may combine or omit internal components while preserving externally relevant behavior.

RAM should determine whether internal structures that are removed could materially affect: failure propagation, dependencies, performance, interaction, safety, or simulation outcomes.

Environmental Abstraction

Simulation environments may abstract: terrain, infrastructure, atmosphere, weather, lighting, electromagnetic conditions, population, traffic, resources, physical obstacles, or other environmental characteristics.

Environmental abstraction should remain appropriate to the scenarios the simulation must support.

Temporal Abstraction

Reality may operate across different timescales.

RAM governs whether temporal processes may be: compressed, aggregated, approximated, event-driven, discretized, or otherwise simplified.

Temporal abstraction should not remove material: latency, accumulation, sequence, delay, degradation, adaptation, feedback, or timing dependencies.

Spatial Abstraction

Spatial reality may be simplified through: reduced geometry, zones, cells, coordinates, topological relationships, representative distances, or other spatial models.

The abstraction must preserve spatial characteristics material to the simulation inquiry.

Interaction Abstraction

Two Reality Elements may be represented accurately individually while their interaction is represented inadequately.

RAM therefore governs abstraction of: coupling, communication, influence, collision, transfer, feedback, synchronization, coordination, and dependency.

Dependency Abstraction

Dependencies may be simplified only where their material effect remains sufficiently represented.

A hidden dependency removed through abstraction can create a simulation that behaves correctly internally while failing to represent real-world system behavior.

Failure Abstraction

Failure behavior often differs from nominal behavior.

Where failure is material to Simulation Purpose, RAM should determine whether the abstraction preserves: failure initiation, failure propagation, degraded states, recovery behavior, cascading effects, and relevant interaction with other elements.

Abstraction Assumption

Every material abstraction may create assumptions.

An Abstraction Assumption states what is presumed to remain sufficiently true despite reduction of real-world complexity.

Example:

Individual variation is assumed not to materially affect the aggregate outcome within the defined scenario range.

Such assumptions should remain explicit and traceable.

Abstraction Justification

A material abstraction should have a governed reason.

Possible reasons may include: characteristic immateriality, known equivalence, validated simplification, bounded approximation, computational necessity, data limitation, scenario limitation, or established domain methodology.

Computational necessity alone does not prove methodological acceptability.

Computational Constraint Rule

SIMULOS® establishes:

A computational limitation may explain an abstraction, but it does not by itself justify that abstraction.

Where computational limits force removal of material characteristics, the resulting limitation must remain visible.

Abstraction Limitation

An Abstraction Limitation identifies where the abstracted representation no longer sufficiently reflects some characteristic of Relevant Reality.

Limitations may constrain: scenarios, operating conditions, interpretation, output applicability, comparison, extrapolation, or operational reliance.

Abstraction Uncertainty

Abstraction may introduce uncertainty because multiple real-world states become represented through one simplified model.

Where material, this uncertainty should remain explicit.

It should not disappear merely because the abstraction itself is deterministic.

Abstraction Sensitivity

Where the acceptability of an abstraction is uncertain, its effect may require sensitivity examination.

The key governance question is:

Would a materially different abstraction plausibly change the simulation outcome or interpretation?

Where yes, the abstraction should receive additional scrutiny.

Abstraction Consistency

Related Reality Elements should not use incompatible abstraction assumptions without explicit justification.

For example, one subsystem should not assume:

instantaneous communication

while another relies upon:

material communication latency

unless the relationship is governed.

Cross-Element Abstraction

Abstraction decisions should therefore be evaluated not only independently but also collectively.

The combined abstraction structure may create effects not visible when each element is reviewed separately.

Abstraction Completeness

Abstraction Completeness asks whether every materially selected Reality Element has received sufficient abstraction treatment to proceed into representation governance.

This does not mean every element must be simplified.

Some may remain detailed.

The requirement is that the abstraction status is known.

Abstraction Status

A selected Reality Element may receive an abstraction status such as:

Detailed Representation Required

Reduced Representation Approved

Aggregated Representation Approved

Parameterized Representation Approved

Approximation Approved

Conditional Abstraction

Abstraction Constraint

or:

Abstraction Undetermined Conditional Abstraction

An abstraction may be acceptable only within defined: scenarios, ranges, environments, time periods, operating modes, populations, or system states.

These conditions should propagate forward into subsequent representation governance.

Abstraction Constraint

An Abstraction Constraint exists where required real-world complexity cannot currently be represented at an acceptable abstraction level.

This should not be silently converted into an approved simplification.

Abstraction Traceability

A governed abstraction should support the chain:

Simulation Purpose → Reality Selection → Reality Element → Material Characteristic → Abstraction Requirement → Abstraction Decision → Abstracted Representation → Limitation

This provides a reconstructable explanation of how reality became simulation representation.

Abstraction Change

Abstraction may require reassessment where: Simulation Purpose changes, Simulation Scope changes, Reality Selection changes, new interactions are identified, new failure modes emerge, representation capability changes, new evidence becomes available, scenario requirements expand, or simulation outputs demonstrate sensitivity to abstraction.

Abstraction Change Propagation

Material abstraction changes may affect:

Fidelity → Resolution → Representation Limitations → Assumptions → Model & Environment → Inputs & Data → Scenario Design → Execution → Outputs → Reality Correlation

Abstraction is therefore an upstream simulation-governance dependency.

Abstraction Basis

The principal governed output of RAM is the Abstraction Basis.

It should establish: Abstraction Objects, material characteristics, abstraction requirements, selected abstraction levels, simplifications, aggregations, decompositions, parameterizations, approximations, abstraction assumptions, abstraction limitations, abstraction uncertainty, abstraction constraints, applicable conditions, and traceability to Reality Selection.

Relationship to Reality Selection

RSM determines:

What reality enters the simulation?

RAM determines:

How may the complexity of that selected reality be reduced?

The distinction protects SIMULOS® from treating omission and abstraction as the same methodological act.

Relationship to Representation Fidelity

RAM does not establish whether the resulting representation achieves the required fidelity.

That is governed by:

RFM — Representation Fidelity Module

RAM provides the abstraction basis against which fidelity can subsequently be evaluated.

Relationship to Representation Resolution

RAM may determine that a phenomenon can be discretized or aggregated.

It does not determine the exact resolution at which representation must operate.

That belongs to:

RRM — Representation Resolution Module Relationship to Representation Limitation

RAM identifies limitations created specifically by abstraction.

RLM — Representation Limitation Module subsequently consolidates and governs the broader limitations of the simulation's Reality Representation.

Cross-Architecture Boundary

RAM does not determine what Operational Reality is.

That remains governed by ORA™.

RAM does not validate whether a simulation may support an external validation claim.

That remains within VALIDOS®.

RAM does not govern general integrity.

That remains within INTEGROS®.

RAM governs one specific transformation:

the controlled reduction of selected reality into simulation-suitable representation.

Minimum Implementation Framework

1. Establish the Reality Selection Basis

Identify the selected Reality Elements and their connection to Simulation Purpose & Scope. 2. Identify Material Characteristics

Determine which properties, behaviors, states, interactions, dependencies, temporal effects, spatial effects, and failure characteristics must survive abstraction. 3. Establish Abstraction Requirements

Define what must remain represented after complexity reduction. 4. Select the Abstraction Approach

Determine whether the element requires detailed, simplified, generalized, aggregated, decomposed, parameterized, discretized, or approximated representation. 5. Evaluate Material Loss

Determine what real-world characteristics are removed, compressed, merged, or approximated. 6. Assess Abstraction Acceptability

Determine whether the resulting representation remains sufficient for the governed simulation inquiry. 7. Record Assumptions, Conditions, and Limitations

Preserve the boundaries under which the abstraction remains legitimate. 8.

Assess Cross-Element Consistency

Determine whether abstraction decisions remain compatible across interacting Reality Elements. 9. Establish the Abstraction Basis

Create the governed representation basis required for fidelity governance. 10. Preserve Traceability and Reassessment

Maintain:

Reality Selection → Material Characteristics → Abstraction → Limitation → Representation

and reassess after material change.

Governance Outputs

RAM may produce: Abstraction Object Register, Material Characteristic Record, Abstraction Requirement, Abstraction Level Record, Simplification Record, Generalization Record, Aggregation Record, Decomposition Record, Parameterization Record, Discretization Record, Approximation Record, Behavioral Abstraction Record, Structural Abstraction Record, Environmental Abstraction Record, Temporal Abstraction Record, Spatial Abstraction Record, Interaction Abstraction Record, Dependency Abstraction Record, Failure Abstraction Record, Abstraction Assumption Register, Abstraction Justification Record, Abstraction Limitation Record, Abstraction Uncertainty Record, Abstraction Sensitivity Record, Abstraction Consistency Assessment, Conditional Abstraction Record, Abstraction Constraint Record, Abstraction Completeness Assessment, Abstraction Change Record, Abstraction Traceability Map, and Abstraction Basis.

Use Case 1 — Autonomous Driving Simulation

Scenario

An autonomous-driving simulation has already selected pedestrians as materially relevant Reality Elements.

Representing every biological and cognitive characteristic of every pedestrian would be unnecessary. Application

RAM determines which characteristics must remain represented, such as: position, movement, direction, speed, reaction variability, visibility, road-crossing behavior, and relevant interaction with the vehicle.

Other characteristics may legitimately remain abstracted where they cannot materially affect the simulation inquiry. Result

The simulation receives a governed pedestrian abstraction that preserves characteristics necessary for the intended safety scenarios without attempting to reproduce the complete human being. Use Case 2 — Spacecraft Thermal Simulation

Scenario

A spacecraft thermal simulation includes the spacecraft structure, solar exposure, internal heat generation, radiation, and relevant environmental conditions. Application

RAM determines whether individual structural components require detailed representation or whether some may be represented through aggregated thermal masses and parameterized heat-transfer relationships.

Where local thermal behavior could materially affect critical components, excessive aggregation is prohibited or recorded as a material limitation. Result

The abstraction remains aligned with the thermal inquiry rather than with computational convenience alone. Use Case 3 — Industrial Digital Twin

Scenario

A digital twin represents a complex industrial production system containing thousands of individual operational components. Application

RAM allows non-material components to be aggregated while preserving individually significant: bottlenecks, dependencies, failure paths, resource constraints, timing relationships, and critical interactions.

Result

The digital twin remains computationally manageable without silently eliminating operational relationships material to the simulation objective.

Architectural Position

Reality Abstraction is the second internal governance space of the Reality Representation Standard (RRS).

The complete RRS progression is:

Reality Selection → Reality Abstraction → Representation Fidelity → Representation Resolution → Representation Limitation

RSM establishes:

What parts of reality matter?

RAM establishes:

How may their complexity be reduced?

RFM establishes:

How faithfully must the resulting representation reproduce the material characteristics required by the simulation?

This sequence prevents three fundamentally different questions from being collapsed into one modeling decision.

Foundational Principle

Abstraction is legitimate only when reducing the complexity of reality does not silently remove the characteristics that make that reality material to the simulation inquiry.

Canonical Closing Statement

Reality Abstraction Module establishes the governed transformation through which selected reality becomes simulation-suitable representation without allowing simplification, aggregation, parameterization, discretization, approximation, or computational constraint to silently remove characteristics material to the simulation inquiry. By governing abstraction requirements, material-characteristic preservation, abstraction levels, assumptions, limitations, uncertainty, cross-element consistency, change propagation, and traceability, RAM ensures that simulation complexity may be reduced where justified while preserving a transparent and reconstructable relationship between the reality selected for simulation and the representation subsequently evaluated for fidelity, resolution, and limitation.

Canonical Source

https://originopenfoundation.org/